
AWS KMS (Key Management Service) — Overview

What is AWS KMS?

AWS Key Management Service (KMS) is a **fully managed encryption service** that allows you to **create, manage, and control cryptographic keys** used to encrypt data.

✅ **Main Goal:** Help you securely encrypt and decrypt your data across AWS services — with minimal effort and maximum control.

Key Concepts

 Component	 Description
 Customer Master Key (CMK)	Main key created in KMS (now known as KMS key); used to encrypt/decrypt data keys or data directly.
 Data Key	Temporary key used for actual encryption of data; encrypted with CMK.
 Envelope Encryption	Secure pattern where data key encrypts the data, and CMK encrypts the data key.
 Key Policy	Controls who can use/manage a key.
 Grants	Temporary access permissions to use keys.
 Automatic Key Rotation	Automatically rotate keys yearly to enhance security.

Types of KMS Keys

Type

Use Case

 Customer Managed Keys (CMKs)	You create, manage, and rotate them. Full control.
 AWS Managed Keys	Automatically created/managed by AWS for each service (e.g., aws/s3).
 AWS Owned Keys	Used internally by AWS; not visible or manageable by customers.

Envelope Encryption (Simplified)

 Instead of encrypting large data directly with CMK (which is slow and costly), KMS does:

1. Generate a **data key**.
2. Encrypt the data with the **data key**.
3. Encrypt the **data key** with a **CMK**.
4. Store encrypted data key + ciphertext together.

 When decrypting:

- Decrypt data key using CMK.
 - Use decrypted data key to decrypt data.
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Supported Services (Examples)

AWS KMS integrates with:

Service

Use Case

 S3	Server-side encryption for buckets (SSE-KMS)
 EBS	Encrypt volumes
 RDS	Encrypt databases
 CloudTrail	Encrypt log files
 Lambda	Encrypt environment variables

KMS Lab (Simple)

 **Objective: Encrypt a file with a KMS key using AWS CLI**

Step 1: Create a CMK

```
aws kms create-key --description "My demo KMS key"
```

Save the **KeyId**.

Step 2: Encrypt a file

```
aws kms encrypt \  
  --key-id <KeyId> \  
  --plaintext fileb://plaintext.txt \  
  --output text \  
  --query CiphertextBlob \  
> encrypted.txt
```

Step 3: Decrypt the file

```
aws kms decrypt \  
  --ciphertext-blob fileb://encrypted.txt \  
  --output text \  
  --query Plaintext | base64 --decode
```

Security & Compliance

- 🔍 **Audit with CloudTrail:** All KMS operations are logged.
 - ↻ **Automatic Key Rotation:** Yearly for CMKs (optional).
 - ⌚ **Grants and key policies** can limit temporary usage.
 - 🧑 **FIPS 140-2 compliance** supported.
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When to Use KMS?

- Need **encryption at rest** or **in transit**.
 - Need **centralized key control** across services.
 - Need **auditable encryption** (logs + access control).
 - Integrating with **custom applications** for secure key handling.
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Summary

 Feature	 Benefit
 CMK	Central control over encryption keys
 Rotation	Automatic yearly key rotation
 Envelope Encryption	Performance + security balance
 Auditing	Integrated with CloudTrail
 Multi-service	Works with S3, RDS, EBS, Lambda, etc.

Bonus: Common Interview Questions

1.  What's the difference between CMK and data key?
 2.  What is envelope encryption?
 3.  How does AWS KMS integrate with S3?
 4.  What is key rotation? Is it enabled by default?
 5.  What happens if I delete a CMK used to encrypt S3 data?
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